

Phasor Representation of Sinusoid

$$v = V_m \cos(\omega t \pm \theta) \Rightarrow \underline{V = V_m \angle \pm \theta}$$

Example 9.4

a) $i = 6 \cos(50t - 40^\circ) \text{ A}$

$$\underline{I = 6 \angle -40^\circ \text{ A}}$$

b) $v = -4 \sin(30t + 50^\circ) \text{ V}$

$$= 4 \cos(30t + 50^\circ + 90^\circ)$$

$$= 4 \cos(30t + 140^\circ)$$

$$\underline{V = 4 \angle 140^\circ \text{ V}}$$

Practice Problem 9.4

Example 9.5

$$\begin{aligned} \text{a) } \mathbf{I} &= -3 + j4 \text{ A} \\ &= 5 \angle 126.87^\circ \text{ A} \\ i &= 5 \cos(\omega t + 126.87^\circ) \text{ A} \end{aligned}$$

Practice Problem 9.5

$$\begin{aligned} \text{a) } \mathbf{V} &= -25 \angle 40^\circ \text{ V} \\ \text{b) } \mathbf{I} &= j(12 - j5) \text{ A} \\ &= j12 - j^2 5 \text{ A} \quad \begin{array}{l} j = \sqrt{-1} \\ j^2 = -1 \end{array} \\ &= j12 + 5 \text{ A} \end{aligned}$$

$$\mathbf{I} = 5 + j12 \text{ A}$$

$$\mathbf{I} = 13 \angle 67.38^\circ \text{ A}$$

$$i = 13 \cos(\omega t + 67.38^\circ) \text{ A}$$

Example 9.6

$$i_1(t) = 4 \cos(\omega t + 30^\circ) \text{ A}$$

Find their sum.

$$\begin{aligned} i_2(t) &= 5 \sin(\omega t - 20^\circ) \text{ A} \\ &= 5 \cos(\omega t - 20^\circ - 90^\circ) \text{ A} \end{aligned}$$

$$i_2(t) = 5 \cos(\omega t - 110^\circ) \text{ A}$$

$$I_1 = 4 \angle 30^\circ \text{ A}$$

$$I_2 = 5 \angle -110^\circ \text{ A}$$

$$\begin{aligned} I &= I_1 + I_2 = 4 \angle 30^\circ + 5 \angle -110^\circ \\ &= 3.218 \angle -56.976^\circ \text{ A} \end{aligned}$$

$$i(t) = 3.218 \cos(\omega t - 56.976^\circ) \text{ A}$$

Practice Problem 9.6

Integration and Differentiation of Phasors

$$\begin{array}{l|l} \frac{dv}{dt} \Leftrightarrow j\omega V & \frac{di}{dt} \Leftrightarrow j\omega I \\ \int v dt \Leftrightarrow \frac{V}{j\omega} & \int i dt \Leftrightarrow \frac{I}{j\omega} \end{array}$$

Example 9.7

$$4i + 8 \int i dt - 3 \frac{di}{dt} = 50 \cos(2t + 75^\circ)$$

$$4I + 8 \frac{I}{j^2} - 3j2I = 50 \angle 75^\circ$$

$$4I + \frac{4I}{j} - j6I = 50 \angle 75^\circ$$

$$4I - j4I - j6I = 50 \angle 75^\circ$$

$$I(4 - j4 - j6) = 50 \angle 75^\circ$$

$$I = \frac{50 \angle 75^\circ}{(4 - j10)} = 4.642 \angle 143.199^\circ$$

$$i = 4.642 \cos(2t + 143.199^\circ) \text{ A}$$

Practice Problem 9.7